



YERDOS NARZHITOV

PROFILE

Strong interest in low-level programming and networking (protocol obfuscation and encryption).
Also interested in UNIX-like environment (especially GNU/Linux).

CONTACT ME



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Gangwon-do,
Chuncheon-si,
Aemakgol-gil



EDUCATION

KANGWON NATIONAL UNIVERSITY

B.Sc. Computer Science and Engineering
2023-2027



LANGUAGE

Advanced English

Native Russian

Native Kazakh

Native Uzbek

Upper-Intermediate Korean (TOPIK 6)



COMPUTER SKILLS

Git, GDB, ASAN, pThreads, LibSodium

C11, Bash, Java, Python

Linux (Arch, Debian, Alma)



WORK EXPERIENCE

OFFICE OF INTERNATIONAL AFFAIRS

2 years - Internation supporter at the Office of International Affairs at Kangwon National University.

CHUNCHEON HONORARY INTERPRETERS

1 year - Translator (English-Russian → Korean) at Chuncheon City Hall.

Engineering Projects

➤ OpenP2P: E2EE Messenger/File-transfer & Proxy (C11, Makefile)

- Peer-to-peer communication tool featuring NAT hole-punching.
- End-to-end encryption and forward secrecy using X25519, XChaCha20-Poly1305.
- Authenticated handshake protocol to resist Man-in-the-Middle (MITM) attacks.
- POSIX threads to handle concurrent operations.
- Validated memory safety using AddressSanitizer (ASAN) and GDB.

➤ MNIST Handwritten Digit Recognition Neural Network (C11)

- Feedforward operation, the backpropagation algorithm, and ReLU activation function through manual memory management in C with no libraries.
- Graphical 28x28 canvas (using SDL2) allows users to draw digits and test the neural network.

➤ 3X-UI Panel Automated Installation (Bash)

- Bash script (>1K LOC) to fully automate the installation of 3X-UI panel.
- Network stack optimization by modifying sysctl configuration file to enable TCP BBR congestion control, and expand TCP buffers, increasing VPN throughput to maximum.
- UFW for firewall, Caddy for reverse proxy and HTTPS/SSL certificate management.

➤ Low-Level PNG Image Processor (C11, Makefile)

- Custom PNG parser from scratch capable of reading, decompressing (via zlib), and validating IHDR, IDAT, IEND chunks, alongside custom hidden chunks manipulation.
- Detects injected hidden data and can inject hidden data chunks (steganography).
- 3x3 convolution kernels to apply image processing filters (Sobel, Gaussian, Laplacian) via direct memory manipulation of pixel matrix.

➤ UDP with ACK Simulation (C11, Bash)

- Simple Stop-and-Wait flow control over UDP with ACK packets.
- Simulates network instability, including packet loss, delays, and retransmission strategies.

➤ Campfire website backend - 모닥불 (Team Project, Python, SQLite3)

- Collaborated in a 4-person agile team utilizing Git version control to build a real-time anonymous community backend.
- Engineered the core topic creation endpoints and implemented a similarity mapping algorithm to track and cluster trending discussions.
- No heavy ORMs to write a highly optimized Raw SQL queries for maximum DB retrieval performance.

Additional Information

➤ Consistent Development Discipline

- Maintained a continuous 6-month daily coding streak (minimum 3 hours/day).
- Over 50 % of the codebase written in C and Bash.

➤ Contributions to open source projects

- Blitz (Obfuscation Protocol: Hysteria Panel): Fixed the installation Bash script.
- Implemented process logging to streamline debugging for broken/interrupted deployments

➤ MAT364 (Security Course of Suleyman Demirel University)

- Corrected the Bash script for Markdown-to-PDF conversion.
- Fixed calculation logic for encryption algorithm.

➤ Server maintenance (VPS)

- Maintained a VPS with 3X-UI running in a docker.
- Added AdGuardHome to block adds by DNS.

➤ Linux Familiarity

- Daily drive 2 Linux Distributions: Arch Linux and Void Linux.
- Main VPS running Debian Linux.
- Currently learning Alma Linux for RHCSA certificate.

➤ PC Build Experience

- Built more than 5 PCs from scratch.
- Familiar with hardware components.
- Experience of preventing bottlenecking by weak parts.